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Bachelor Of Technology
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EDUCATION

•National Institute of Technology, Raipur	Aug.2024-Present
B-Tech in Civil Engineering	7.22
•Shanti Niketan Academy, Gaya, Bihar	2022-23
Central Board of Secondary Education	85.8
•D.A.V. Public School Cantt. Area Gaya, Bihar	2020-21
Central Board of Secondary Education	88.8

EXPERIENCE

•INDIAN NATIONAL SCIENCE ACADEMY RESEARCH FELLOW (IIT BOMBAY)	May 2026 - JUL 2026
INSA FELLOW at IIT BOMBAY	
– Developed a CDR-based machine learning model using mobile phone Call Detail Records (CDRs) to estimate urban travel demand, reducing dependence on expensive household travel surveys.	
– Designed a hybrid Population Synthesis pipeline by integrating Variational Autoencoders (VAE) with Iterative Proportional Updating (IPU) to generate realistic census-scale synthetic populations from limited survey data.	
– Generated diverse synthetic population profiles using a VAE and recalibrated them with IPU to preserve demographic relationships while ensuring consistency with official census statistics for agent-based transportation microsimulation.	

PERSONAL PROJECTS

–Embrya — Embryo Viability Analysis Platform	DEC. 2025
Built a full-stack clinical-decision-support platform that scores IVF embryo viability from microscope images.	
* Designed an ensemble machine learning pipeline of three SMOTE-balanced Random Forest classifiers to predict embryo viability from extracted morphological features.	
* Built an OpenCV-based feature extraction engine computing circularity, symmetry, fragmentation, edge density, and entropy from embryo images for model inference.	
* Engineered a FastAPI backend exposing a REST prediction endpoint with request validation, feature scaling, and probability-averaged ensemble inference.	
* Developed a React + TypeScript clinical dashboard (Tailwind CSS, Radix UI) with real-time scoring, batch processing, embryo comparison, and PDF/TXT report generation.	
–IR Image Colorization & Enhancement	JUL. 2026
Satellite infrared-to-RGB translation system using super-resolution, semantic segmentation and conditional image synthesis.	
* Built a 4-stage pipeline (Real-ESRGAN → DeepLabV3+ → Pix2PixHD → Semantic Correction) to colorize Landsat 8/9 infrared satellite tiles into realistic RGB imagery, trained on 4,940 paired IR/RGB tiles from USGS.	
* Trained a DeepLabV3+ segmentation model (ResNet-101 backbone, adapted to single-channel IR input) using auto-generated pseudo-labels via spectral indices (ExG, NDWI), eliminating the need for manual annotation.	
* Designed a 4-channel conditioned Pix2PixHD generator (IR + one-hot segmentation map) with a multi-scale PatchGAN discriminator, achieving semantically consistent colorization with explicit land-cover class awareness.	
* Deployed an end-to-end inference dashboard (Gradio) enabling upload of a raw IR image and real-time output of the full enhance → segment → colorize pipeline result.	

TECHNICAL SKILLS AND INTERESTS

Programming Languages: Python, JavaScript, HTML, CSS
Data Analytics: Data Analysis, Microsoft Excel, SQL, NumPy, Pandas
Machine Learning: Machine Learning
Developer Tools: VS Code, Jupyter Notebook
Soft Skills: Analytical Thinking, Problem Solving, Process Optimization, Teamwork, Leadership, Communication
Areas of Interest: Artificial Intelligence, Data Analytics, Transportation Engineering

POSITIONS OF RESPONSIBILITY

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| – Executive , Civil Engineering Association (CEA) | <i>JUL..2025-Present</i> |
| – Coordinator , National Service Scheme (NSS) | <i>OCT.2024-Present</i> |

ACHIEVEMENTS

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| – Amazon ML Summer School 2026 Scholar , Amazon | <i>JUL. 2026</i> |
| – INSA Summer Research Fellow , Indian National Science Academy (INSA) | <i>MAY. 2026</i> |
| – National Finalist , Smart India Hackathon (SIH) | <i>DEC. 2025</i> |
| – Selected Intern , NHA Winter Paid Internship | <i>DEC. 2025</i> |
| – Runner-up , Truss Bridge Making Competition, Civil Engineering Department, NIT Raipur | <i>FEB. 2025</i> |
| – Runner-up , Mad Over Marketing – Business Case Study Competition, E-Cell NIT Raipur | <i>MAR. 2025</i> |
| – National Level Merit Holder , Indian Olympiad Qualifier in Mathematics (IOQM) | <i>SEPT. 2020</i> |
| – National Level Merit Holder, AIR (15) , Indian National Mathematics Talent Contest | <i>NOV. 2018</i> |